

COMPUTER GRAPHICS

Unit-3: 2D Transformations and Viewing

Comprehensive Study Notes

BCA Semester VI - HPU

2D Transformations and Viewing

Introduction to 2D Transformations

Definition: 2D Transformations are operations that change the position, size, orientation, or shape of objects in a two-dimensional space. These transformations are fundamental to computer graphics for manipulating objects and creating animations.

Why Homogeneous Coordinates?

Definition: Representation of 2D points as 3D vectors.

- Normal Coordinates: (x, y)
- Homogeneous: $(x, y, 1)$ or (x, y, w)

Why Use Homogeneous Coordinates?

1. Allows representation of translation as matrix operation
2. All transformations become matrix operations
3. Composite transformations become matrix multiplication
4. Points at infinity can be represented
5. Consistent mathematical framework

Point Representation: (x, y, w) where $w \neq 0$. Actual point: $(x/w, y/w)$. For 2D: $w = 1 \rightarrow (x, y, 1)$.

1. Translation

Definition: Translation is a transformation that moves an object from one position to another without changing its size, shape, or orientation.

Mathematical Representation:

- Original Point: (x, y)
- Translation: (t_x, t_y)
- New Point: $(x', y') = (x + t_x, y + t_y)$

Matrix Form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x+t_x \\ y+t_y \\ 1 \end{bmatrix}$$

Properties:

1. Preserves shape and size
2. Preserves orientation
3. Commutative: $T_1 + T_2 = T_2 + T_1$
4. Additive: $T(t_{x1}, t_{y1}) + T(t_{x2}, t_{y2}) = T(t_{x1}+t_{x2}, t_{y1}+t_{y2})$

Translation Example - Triangle:

Given Triangle: $A(2,2)$, $B(4,2)$, $C(3,4)$
 Translation: $t_x = 3$, $t_y = 2$

Calculation:

$$A'(2+3, 2+2) = A'(5, 4)$$

$$B'(4+3, 2+2) = B'(7, 4)$$

$$C'(3+3, 4+2) = C'(6, 6)$$

2. Rotation

Definition: Rotation is a transformation that turns an object about a fixed point (pivot point) by a specified angle.

Mathematical Representation:

- Original Point: (x, y)
- Rotation Angle: θ
- $x' = x \cdot \cos\theta - y \cdot \sin\theta$
- $y' = x \cdot \sin\theta + y \cdot \cos\theta$

Matrix Form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotation About Arbitrary Point:

1. Translate to origin: $T(-x_p, -y_p)$
2. Rotate about origin: $R(\theta)$
3. Translate back: $T(x_p, y_p)$

$$P'(x', y') = T(x_p, y_p) \times R(\theta) \times T(-x_p, -y_p) \times P$$

Properties:

1. Preserves shape and size
2. Changes orientation
3. Not commutative: $R(\theta_1) \cdot R(\theta_2) \neq R(\theta_2) \cdot R(\theta_1)$
4. Inverse: $R(-\theta) = R(\theta)^{-1}$

Rotation Example - Triangle (45° about Origin):

Given Triangle: $A(2, 1)$, $B(5, 1)$, $C(3, 4)$
 Rotation: $\theta = 45^\circ$ ($\pi/4$)
 $\cos 45^\circ = \sin 45^\circ = 0.707$

$$\begin{aligned} A'(2, 1): \quad x' &= 2 \times 0.707 - 1 \times 0.707 = 0.707 \sim 1 \\ y' &= 2 \times 0.707 + 1 \times 0.707 = 2.121 \sim 2 \\ A' &= (1, 2) \end{aligned}$$

$$B'(5, 1): \quad x' = 5 \times 0.707 - 1 \times 0.707 = 2.828 \sim 3$$

$$y' = 5 \times 0.707 + 1 \times 0.707 = 4.242 \sim 4$$

$$B' = (3, 4)$$

$$C' (3, 4): x' = 3 \times 0.707 - 4 \times 0.707 = -0.707 \sim -1$$

$$y' = 3 \times 0.707 + 4 \times 0.707 = 4.949 \sim 5$$

$$C' = (-1, 5)$$

Result: $A' (1, 2)$, $B' (3, 4)$, $C' (-1, 5)$

3. Scaling

Definition: Scaling is a transformation that changes the size of an object by multiplying coordinates by scaling factors.

Mathematical Representation:

- Original Point: (x, y)
- Scale Factors: s_x, s_y
- New Point: $(x', y') = (x \cdot s_x, y \cdot s_y)$

Matrix Form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \cdot s_x \\ y \cdot s_y \\ 1 \end{bmatrix}$$

Scaling Effects:

Condition	Effect
$s_x > 1$	Scale up (expand) in x-direction
$s_x < 1$	Scale down (compress) in x-direction
$s_y > 1$	Scale up (expand) in y-direction
$s_y < 1$	Scale down (compress) in y-direction
$s_x = s_y$	Uniform scaling
$s_x \neq s_y$	Differential scaling

Scaling About Fixed Point:

1. Translate to origin: $T(-x_f, -y_f)$
2. Scale: $S(s_x, s_y)$
3. Translate back: $T(x_f, y_f)$

$$P'(x', y') = T(x_f, y_f) \times S(s_x, s_y) \times T(-x_f, -y_f)$$

Scaling Example - Rectangle:

Given Rectangle: $A(1,1)$, $B(4,1)$, $C(4,3)$, $D(1,3)$
Scale Factors: $s_x = 2$, $s_y = 1.5$

Calculation:

$$A'(1 \times 2, 1 \times 1.5) = A'(2, 1.5)$$

$$B'(4 \times 2, 1 \times 1.5) = B'(8, 1.5)$$

$$C'(4 \times 2, 3 \times 1.5) = C'(8, 4.5)$$

$$D'(1 \times 2, 3 \times 1.5) = D'(2, 4.5)$$

Result: $A'(2,1.5)$, $B'(8,1.5)$, $C'(8,4.5)$, $D'(2,4.5)$

4. Reflection

Definition: Reflection (also called Mirroring) is a transformation that produces a mirror image of an object across a line or axis.

Types of Reflection:

Type	Equations	Matrix
About X-axis	$x' = x, y' = -y$	$[1 \ 0 \ 0; 0 \ -1 \ 0; 0 \ 0 \ 1]$
About Y-axis	$x' = -x, y' = y$	$[-1 \ 0 \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1]$
About Origin	$x' = -x, y' = -y$	$[-1 \ 0 \ 0; 0 \ -1 \ 0; 0 \ 0 \ 1]$
About line $y = x$	$x' = y, y' = x$	$[0 \ 1 \ 0; 1 \ 0 \ 0; 0 \ 0 \ 1]$
About line $y = -x$	$x' = -y, y' = -x$	$[0 \ -1 \ 0; -1 \ 0 \ 0; 0 \ 0 \ 1]$

Reflection Example - Triangle:

Given Triangle: A(2,2), B(5,2), C(3,5)

1. About X-axis: A'(2, -2) B'(5, -2) C'(3, -5)
2. About Y-axis: A'(-2, 2) B'(-5, 2) C'(-3, 5)
3. About Origin: A'(-2, -2) B'(-5, -2) C'(-3, -5)
4. About line $y = x$: A'(2, 2) B'(2, 5) C'(5, 3)

5. Shearing

Definition: Shearing is a transformation that slants or skews an object by shifting coordinates in one direction proportional to the other coordinate.

Mathematical Representation:

Type	Equations	Matrix
X-Shear	$x' = x + sh_x \cdot y, y' = y$	$[1 \ sh_x \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1]$
Y-Shear	$x' = x, y' = sh_y \cdot x + y$	$[1 \ 0 \ 0; sh_y \ 1 \ 0; 0 \ 0 \ 1]$

Properties:

1. Changes shape of object
2. Area preserved (determinant = 1)
3. Not commutative with rotation
4. Used in creating 3D effects from 2D
5. Used in font design (italic text)

Shearing Example - Rectangle:

Given Rectangle: A(1,1), B(4,1), C(4,3), D(1,3)
 X-Shear: $sh_x = 1.5$, Y-Shear: $sh_y = 0.5$

X-Shear Calculation:

$A'(1 + 1.5 \times 1, 1) = A'(2.5, 1)$
 $B'(4 + 1.5 \times 1, 1) = B'(5.5, 1)$
 $C'(4 + 1.5 \times 3, 3) = C'(8.5, 3)$
 $D'(1 + 1.5 \times 3, 3) = D'(5.5, 3)$

Y-Shear Calculation:

$A'(1, 0.5 \times 1 + 1) = A'(1, 1.5)$
 $B'(4, 0.5 \times 4 + 1) = B'(4, 3)$
 $C'(4, 0.5 \times 4 + 3) = C'(4, 5)$
 $D'(1, 0.5 \times 1 + 3) = D'(1, 3.5)$

6. Composite Transformation

Definition: Composite Transformation is the combination of multiple basic transformations applied in sequence to achieve a complex result.

Mathematical Representation:

If transformations are: $T_1, T_2, T_3, \dots$
 Applied in order: T_1 then T_2 then T_3
 Composite Matrix: $M = T_3 \times T_2 \times T_1$
 Note: Order matters! Matrices multiplied from right to left

Example: Rotation about arbitrary point

$$M = T(x_p, y_p) \times R(\theta) \times T(-x_p, -y_p)$$

Example: Scaling about fixed point

$$M = T(x_f, y_f) \times S(s_x, s_y) \times T(-x_f, -y_f)$$

Composite Transformation Example - Rotation about Point:

Given: Rotate triangle about point $(2, 2)$ by 90°
 Triangle: $A(1, 1), B(3, 1), C(2, 3)$

Step 1: Translate to origin: $T(-2, -2)$
 Step 2: Rotate 90° : $R(90^\circ)$
 Step 3: Translate back: $T(2, 2)$

Composite: $M = T(2, 2) \times R(90^\circ) \times T(-2, -2)$

$A(1, 1)$: $T(-2, -2)$: $(-1, -1)$
 $R(90^\circ)$: $(1, -1)$
 $T(2, 2)$: $(3, 1) \Rightarrow A' = (3, 1)$

$B(3, 1)$: $T(-2, -2)$: $(1, -1)$
 $R(90^\circ)$: $(1, 1)$
 $T(2, 2)$: $(3, 3) \Rightarrow B' = (3, 3)$

$C(2, 3)$: $T(-2, -2)$: $(0, 1)$
 $R(90^\circ)$: $(-1, 0)$
 $T(2, 2)$: $(1, 2) \Rightarrow C' = (1, 2)$

Result: $A'(3, 1), B'(3, 3), C'(1, 2)$

7. Two Dimensional Viewing

7.1 Window-to-Viewport Coordinate Transformation

Definition: Window-to-Viewport transformation is the process of mapping a rectangular region in world coordinates (window) to a rectangular region in device coordinates (viewport).

Definitions:

Concept	Description
Window (World Coordinates)	Rectangular region in world space defining what part of scene to view
Viewport (Device Coordinates)	Rectangular region on display device defining where to display the window

Transformation Formula:

```
Scaling Factors:
sx = (xv_max - xv_min) / (xw_max - xw_min)
sy = (yv_max - yv_min) / (yw_max - yw_min)

Mapping:
xv = xv_min + (xw - xw_min) x sx
yv = yv_min + (yw - yw_min) x sy

Matrix Form:
|xv|   |sx  0  -sx·xw_min| |xw|
|yv| = |0  sy -sy·yw_min| |yw|
|1 |   |0   0         1   | |1 |
```

Window-to-Viewport Example:

```
Given:
Window: xw_min=0, xw_max=10, yw_min=0, yw_max=8
Viewport: xv_min=50, xv_max=450, yv_min=50, yv_max=350
Point in Window: P(3, 4)

Step 1: Calculate scaling factors
sx = (450-50)/(10-0) = 400/10 = 40
sy = (350-50)/(8-0) = 300/8 = 37.5

Step 2: Apply transformation
xv = 50 + (3-0)x40 = 50 + 120 = 170
```

$$y_v = 50 + (4-0) \times 37.5 = 50 + 150 = 200$$

Result: P'(170, 200) in viewport coordinates

Quick Reference Summary - Unit III

Key Terms:

1. **Translation** - Moving object by (t_x, t_y)
2. **Rotation** - Turning object by angle θ
3. **Scaling** - Resizing object by (s_x, s_y)
4. **Reflection** - Mirroring object about axis
5. **Shearing** - Slanting/Skewing object
6. **Composite** - Multiple transformations combined
7. **Homogeneous Coordinates** - $(x, y, 1)$ representation
8. **Window** - World coordinate region
9. **Viewport** - Device coordinate region
10. **World-to-Screen** - Window-to-Viewport mapping

Transformation Matrices Quick Reference:

```

Translation:  [1 0 tx; 0 1 ty; 0 0 1]
Rotation:     [cosθ -sinθ 0; sinθ cosθ 0; 0 0 1]
Scaling:      [sx 0 0; 0 sy 0; 0 0 1]
X-Shear:      [1 shx 0; 0 1 0; 0 0 1]
Y-Shear:      [1 0 0; shy 1 0; 0 0 1]
X-Reflection: [-1 0 0; 0 1 0; 0 0 1]
Y-Reflection: [1 0 0; 0 -1 0; 0 0 1]
Origin Ref:   [-1 0 0; 0 -1 0; 0 0 1]
  
```

2D Transformations Summary Table:

Transform	Equation	Matrix
Translation	$x' = x + t_x, y' = y + t_y$	$[1 \ 0 \ t_x; 0 \ 1 \ t_y; 0 \ 0 \ 1]$
Rotation (θ)	$x' = x \cdot \cos\theta - y \cdot \sin\theta$ $y' = x \cdot \sin\theta + y \cdot \cos\theta$	$[\cos\theta \ -\sin\theta \ 0; \sin\theta \ \cos\theta \ 0; 0 \ 0 \ 1]$
Scaling	$x' = x \cdot s_x, y' = y \cdot s_y$	$[s_x \ 0 \ 0; 0 \ s_y \ 0; 0 \ 0 \ 1]$
X-Shear	$x' = x + sh_x \cdot y, y' = y$	$[1 \ sh_x \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1]$

Y-Shear	$x'=x, y'=sh_y \cdot x+y$	$[1 \ 0 \ 0; sh_y \ 1 \ 0; 0 \ 0 \ 1]$
Reflection (Y-axis)	$x'=-x, y'=y$	$[-1 \ 0 \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1]$
Reflection (X-axis)	$x'=x, y'=-y$	$[1 \ 0 \ 0; 0 \ -1 \ 0; 0 \ 0 \ 1]$

Review Questions - Unit III

1. What are homogeneous coordinates? Why are they used in computer graphics?
2. Derive the matrix representation for 2D translation, rotation, and scaling.
3. Explain the difference between uniform and differential scaling.
4. Derive the composite transformation matrix for rotation about an arbitrary point.
5. What is reflection? Write the matrices for reflection about X-axis, Y-axis, and origin.
6. Explain shearing transformation with examples for both X-shear and Y-shear.
7. What is the difference between window and viewport? Explain the window-to-viewport transformation.
8. Prove that translation transformations are commutative.
9. Explain the importance of order in composite transformations with an example.
10. Write a short note on the applications of 2D transformations in computer graphics.